

SBC 201 Chapter 16 Structural Design — Group R-2 High-Rise Dimensional Requirements Matrix (2024)

1. Document metadata and use limitation

- **Project basis:** Riyadh, Saudi Arabia; Group R-2 residential high-rise; an occupied floor is stated to be more than 23 m above the relevant reference level.
- **Deliverable tier:** Project-use matrices in Sections 1–19 (design-check rows, not pasted inventory), plus a coverage summary and unresolved-source register. The full row inventory is not published.
- **Code/source basis:** SBC 201 (2024), Chapter 16, source file Reference\SBC 201 2024\source_reference\Chapter_16 – STRUCTURAL DESIGN.txt.
- **Extraction audit:** Skill extract. Project-use rows follow the chapter-extract row contract (noun-phrase checks, bold SI values, building-language triggers, named exceptions, check-specific actions). Internal inventory: **437** independently checkable numeric records (**191** Verified, **246** Verify source). Unresolved OCR is listed in the register and is not a design-release value.
- **Model:** Cursor Grok 4.6.
- **Prepared:** 2026-08-27.
- **Status:** Source-only architectural/structural-coordination advisory matrix. It is not a stamped structural calculation package, geotechnical report, SCD NOC, SBPS approval, legal opinion, or permission to construct.
- **Outbound-source rule:** No value in this matrix has been imported from SBC 301 (including mapped wind or seismic figures), SBC 303–308, SBC 701, SBC 801, ICC 600, ICC 500, ICC 300, AWC WFCM, AISI S230, ASME A17.1, AA ADM, commentary examples, or the existing chapter summary. Where Chapter 16 sends the user elsewhere, this matrix records the dependency without supplying the outbound value.

Scope and assumptions

1. Group R-2 and high-rise status are project statements, not independently verified classifications. Chapter 16 uses “high-rise” in Section 1616 without restating a height trigger.
2. The exact Riyadh AHJ/permit pathway, project stage, SCD NOC status and structural engineer of record are unconfirmed; therefore this matrix does not conclude compliance.
3. Risk category is unconfirmed. A conventional R-2 apartment tower that is not a listed essential facility typically sits in Table 1604.5 outside Categories I, III and IV, but occupant load > **5,000**, mixed assembly, and shared egress with a higher-category occupancy can raise the category. Flattened Table 1604.5 cells are **Verify source**.
4. Automatic sprinkler protection is not a Chapter 16 live-load branch. Do not assume NFPA 13 versus 13R from this extract.
5. Building height, storey count, site class, wind exposure, flood-hazard status, parking/garage type, occupiable roof, mixed-use podium, BMU/fall-arrest provision and vegetative/PV roofs are unconfirmed.
6. All appended load, deflection, wind-conversion, soil and seismic-coefficient tables are concatenated OCR. No reconstructed table cell is adopted as a design-release value.
7. Snow load (1603.1.3, 1608.1) and atmospheric ice load (1614.1) are stated inapplicable to KSA.

2. Legends

Applicability

Status	Meaning
Direct	Expected to govern the stated R-2 tower basis, subject to confirmed geometry and design data.
Conditional	Governs only when the stated feature, use, occupant load, mixed-use/podium condition or structural system exists.
Not typical	Unrelated occupancy-only rule; omitted from this deliverable unless the gap register already opened that use.
External verification	Chapter 16 points to another section/code/standard, or the project/AHJ basis must be confirmed before use.

Source confidence

Status	Meaning
Verified	Requirement and any stated numeric value were checked against unambiguous mandatory Chapter 16 source text.
Verify source	OCR, flattened table, page-split, or footnote attachment is unresolved. Not a design-release value.

3. Project decision and gap register

Decision / gap	Current project basis	Why it controls Chapter 16 application	Required project action
Risk category	Unconfirmed; conventional R-2 is typically II unless a Table 1604.5 III/IV trigger applies	Sets wind-map figure, seismic importance data on CDs, SDC path and whether 1616 integrity applies	Freeze occupancy and occupant load (net area permitted by Table 1604.5 Note a); do not assume II if OL or mixed use trips III
Occupant load vs 5,000	Unconfirmed	Table 1604.5 lists any occupancy with occupant load greater than 5,000 as Risk Category III (flattened OCR)	Issue a signed occupant-load schedule; if the tower or mixed building exceeds the published threshold, design as III
Mixed-use podium / assembly	Unconfirmed	1604.5.1 takes the highest risk category; shared egress/life-safety with a higher-category portion pulls both up	Freeze occupancy by room; structurally separate or design the whole for the higher category
Site class / geotechnical	Unconfirmed	Unknown soils default to Site Class D with Fa not less than 1.2 (1613.2.2–1613.2.3); Table 1610.1 is OCR	Obtain SBC 303 geotechnical data before locking SDC, earth pressure and bearing values
Wind exposure / basic speed	Unconfirmed; maps are figures not readable as values here	1609.3 sends V to Figures 1609.3(1)–(4) or SBC 301; Exposure B/C/D uses 9 m / 450 m / 780 m / 1500 m / 20h tests	SER to lock V, exposure and internal pressure on CDs; do not invent a Riyadh map speed
Occupiable / vegetative / PV roof	Unconfirmed	Occupiable roofs use occupancy-served live load (Table 1607.1 OCR); vegetative soil is dead load wet and dry (1606.5); PV has 600 mm live-load omission	State roof use on architectural plans; do not design an amenity roof as ordinary 1 kN/m² maintenance load without a published-table check
Parking / fire-truck access	Unconfirmed	Passenger-garage concentrated 13.5 kN / 115 mm × 115 mm ; heavy vehicles > 45 kN GVWR and > 4500 kg garages take roadway loads	Confirm podium/basement parking class and SCD appliance routes; post vehicle-weight limits if 1607.8 applies
BMU / fall arrest	High-rise façade access is expected, not independently verified	Hoist supports 2.5× rated/stall load; lifeline anchorages 13.8 kN per line	Show davit/anchor layout and load path on structural and façade drawings

Decision / gap	Current project basis	Why it controls Chapter 16 application	Required project action
Flood / tsunami	Inland Riyadh basis; flood hazard unconfirmed	1612 and 1615 send to SBC 301; 1603.1.7 CD data only if in a flood hazard area	Confirm AHJ flood-hazard map; do not apply coastal tsunami rules without a designated zone
Structural integrity (1616)	High-rise stated; risk category unconfirmed	1616 applies only to high-rise and Risk Category III or IV	If the building remains II, do not treat 1616 ties as a charging requirement; if III/IV, apply 1616.2 or 1616.3
NOC / SER	Unconfirmed	Wind, seismic, soil and integrity values live in SBC 301 / SBC 303–308	Engage the structural engineer of record and geotechnical engineer before design freeze

4. Scope, construction documents and outbound companions

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1601.1	Chapter scope	This chapter governs structural design of buildings, structures and portions thereof regulated by this code	All buildings in scope	None stated	Direct	Treat Chapter 16 as the loading and analysis chapter; materials remain in later chapters	Verified
1603.1	Construction-document content	Show size, section and relative locations of structural members with floor levels, column centres and offsets fully dimensioned; show design loads required by 1603.1.1–1603.1.9	Construction documents	Architectural dimensions need not be duplicated on structural sheets	Direct	Put a structural load-criteria sheet on the CD set	Verified
1603.1.1–1603.1.2	Floor and roof live-load notation	Indicate uniform, concentrated and impact floor live loads by area; indicate roof live load by roof area; show live-load reduction where used (1607.12 / 1607.14)	Floor and roof design	None stated	Direct	Schedule live loads by room use, including any reduction taken	Verified
1603.1.3 / 1608.1	Roof snow load	Roof snow load data and snow load are inapplicable to KSA	KSA projects	Do not import snow combinations from other codes	Direct	Omit snow from the load-criteria sheet	Verified
1603.1.4	Wind data on CDs	Show V (m/s), Vasd , risk category, exposure (and direction if more than one), internal pressure coefficient, and C&C design wind pressures with zone dimensions (kN/m²)	All structures, even if wind does not govern the LFRS	None stated	Direct	Complete all five wind items on CDs; do not leave C&C pressures to the façade contractor without zones	Verified
1603.1.5	Earthquake data on CDs	Show risk category, Ie , Ss , S1 , site class, SDS , SD1 , seismic design category, basic SFRS, design base shear(s), Cs , R , and analysis procedure	All structures, even if seismic does not govern the LFRS	1613.1 exceptions (not this high-rise R-2)	Direct	Complete all eleven seismic items; values come from 1613 / SBC 301, not from this row	Verified
1603.1.6	Soil bearing on CDs	Show design load-bearing values of soils	Foundation design	None stated	Direct	Transfer geotechnical allowable/ultimate bearing onto the foundation notes	Verified
1603.1.8.1 / 1606.4	Rooftop PV dead load	Indicate rooftop PV panel system dead load including rack supports; treat PV, supports and ballast as dead load	Rooftop PV present	None stated	Conditional	If PV is proposed, show saturated array weight and ballast on the roof plan	Verified
1603.1.9	Rain intensity on CDs	Show rain intensity i (cm/hr) whether or not rain governs	Roofs	None stated	Direct	Put design rainfall intensity on the roof-drainage / structural notes	Verified

5. Risk category and mixed occupancy

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1604.5	Risk-category assignment	Assign each building a risk category in accordance with Table 1604.5; do not take a referenced-standard occupancy category lower; use Table 1604.5 in lieu of SBC 301 Table 1.5-1	All buildings	Tsunami Risk Categories III and IV may follow SBC 301 §6.4	Direct	Lock one risk category on the load-criteria sheet before wind/seismic maps are read	Verified
Table 1604.5	Category II default vs III/IV triggers	Flattened OCR table. Category II is buildings not listed in I, III or IV. Category III includes (among other listed uses) any occupancy with occupant load greater than 5,000 , and listed assembly/education/care thresholds. No reconstructed cell is adopted	Occupancy and occupant load	Note a permits net floor area where Table 1004.5 would require gross. Note b (SBC 301 §1.6.3) may drop toxic/explosive III/IV to II if approved	Direct	Use Category II only after confirming the tower is not a listed III/IV use; verify the published 5,000 and assembly 300 / 2,500 cells before applying them	Verify source
1604.5.1	Mixed-occupancy risk	Two or more occupancies not in the same category → assign the highest . Structurally separated portions may be classified separately unless a separated portion provides required access, required egress or shared life-safety components with a higher-category portion — then both take the higher	Mixed-use or structurally separated wings	ICC 500 storm shelter inside a host keeps the host occupancy unless the shelter is a designated emergency shelter in Table 1604.5	Conditional	If a podium assembly/clinic shares egress with the tower, raise the tower to that category or fully separate structure and egress	Verified

6. Serviceability, deflection and glass framing

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1604.3 / Table 1604.3	Member deflection limits	Structural systems shall have adequate stiffness to limit deflections as indicated in Table 1604.3. The table is concatenated OCR (L/360, L/240, L/180, L/120 and finish-dependent wall/partition rows). No reconstructed ratio is adopted	Members listed in the table	More restrictive limits in 1604.3.2–1604.3.5 material standards govern when tighter. Footnotes a, d, f, h, i are flattened with the table	Direct	Specify finish type (plaster/stucco, other brittle, flexible) on walls and partitions so the SER can pick the matching row after a published-table check	Verify source
1604.3.2–1604.3.5	Material deflection standards	Concrete → SBC 304; steel → SBC 306 / 308 / ASCE 8 / SJI; masonry → SBC 305; aluminium → AA ADM	Those materials	Use the more restrictive of the standard and Table 1604.3 (1604.3.6)	External verification	Do not import millimetre or L/n limits from those standards here	Verified
1604.3.7	Glass-supporting framing deflection	Under 0.6 times component-and-cladding wind loads: not more than 1/175 of span for framing not more than 4100 mm ; 1/240 of span + 6.4 mm for framing greater than 4100 mm	Framing members supporting glass	None stated	Direct	Check curtain-wall and window-wall mullion spans against the 4100 mm break and the two limits	Verified
1604.4	Rigid-diaphragm test	A diaphragm is rigid for story-shear and torsion distribution when its lateral deformation is less than or equal to two times the average story drift	Lateral analysis	SBC 301 may still require torsion from eccentricity	External verification	SER to classify each floor diaphragm; architects should not assume a rigid plate	Verified

7. Anchorage, decks and seismic detailing

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1604.8.1	Roof-wall-foundation anchorage	Anchor roof to walls and columns, and walls and columns to foundations, to resist uplift and sliding from the prescribed loads	All structures	None stated	Direct	Show a complete vertical and lateral load path on typical details	Verified
1604.8.2	Structural-wall anchorage	Load-bearing or shear walls shall be anchored to the roof and to all floors and members that support or are supported by the wall, for SBC 301 §1.4.4 (SDC A) or §12.11 (other SDCs). Hollow-unit/cavity masonry anchors shall be embedded in a reinforced grouted structural element	Structural walls	Wind 1609 and earthquake 1613 still apply	External verification	Detail positive wall-to-diaphragm ties; do not import SBC 301 anchorage forces here	Verified
1604.8.3	Deck positive anchorage	Decks supported by attachment to an exterior wall shall be positively anchored for vertical and lateral loads; toenails or nails subject to withdrawal are not permitted . If positive connection cannot be verified in inspection, the deck shall be self-supporting. Cantilevered deck connections shall be designed for full-deck D+L and for live load on the cantilever only	Exterior decks attached to the wall	None stated	Conditional	If unit or amenity decks hang from the façade, use mechanical positive anchors and the two cantilever load cases	Verified
1604.9	Seismic detailing if wind governs	LFRS shall meet seismic detailing in this code and SBC 301 Chapters 11, 12, 13, 15, 17 and 18 even where wind effects exceed seismic	Lateral force-resisting system	SBC 301 references to Chapter 14 shall not apply except as specifically required herein	Direct	Do not drop seismic detailing because wind drift or strength governs	Verified
1604.10	Storm-shelter loads	Storm-shelter loads and combinations shall be determined in accordance with ICC 500	Storm shelters	Do not substitute Risk Category IV wind speeds for ICC 500	Conditional	If a shelter is provided, use ICC 500; do not import those speeds here	External verification

8. Load combinations and dead load

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1605.1	Strength / ASD combinations	Design for SBC 301 §2.3 strength, SBC 301 §2.4 ASD, or 1605.2 alternative ASD, including SBC 301 Chapters 18 and 19 modifications	All structures	ASD: roof live 1.45 kN/m² or less need not be combined with seismic. ASD: crane hook need not combine with roof live or one-half of wind	External verification	SER to select the combination set; ordinary roof live at or below 1.45 kN/m² may omit seismic coincidence under ASD	Verified
1605.2	Alternative ASD combinations	Permitted in lieu of SBC 301 §2.4. Counteracting dead+wind uses only two-thirds of the minimum dead load likely in place. Combinations include 0.6W, E/1.4 and 0.9D + E/1.4 . Foundation overturning reduction SBC 301 §12.13.4 shall not be used with these combinations	Designer elects 1605.2	Crane hook vs roof live / half wind; roof live 1.45 kN/m² or less vs seismic	Conditional	If this set is used, apply the 2/3 dead-load cap on wind uplift cases	Verified
1606.2	Construction-material weights	Use actual material weights; if unknown, values are subject to building-official approval	Dead-load take-off	Outbound unit weights live in SBC 301 Tables 3-1 and 3-2 (not imported)	Direct	Base superimposed dead load on the architectural finish/MEP schedule, not on assumed catalogues	Verified
1606.3	Fixed-equipment weight	Include empty weight plus maximum contents; variable contents (liquids, movable trays) shall not counteract overturning, sliding or uplift per SBC 301 §1.3.6	Fixed service equipment	(1) Variable contents may counteract force effects they themselves cause, with present and absent cases. (2) Seismic force effects need not exceed contents expected in normal operation	Direct	Show tank/riser/equipment operating vs empty cases on the structural notes	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1606.5	Vegetative-roof dead load	Landscaping and hardscaping shall be dead load, computed for fully saturated and fully dry soil and drainage-layer materials, using the more severe effect	Vegetative or landscaped roofs	Live load of the occupiable surface is separate (1607.14.2.2 / Table 1607.1)	Conditional	If a green roof or roof garden is planned, schedule wet and dry soil weights	Verified

9. Live loads for R-2 uses

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1607.3	Minimum uniform live load	Design live load is the maximum expected by intended use, but not less than Table 1607.1	Occupied floors and similar surfaces	Table 1607.1 is four-page concatenated OCR; no reconstructed occupancy cell is adopted	Direct	Assign a published Table 1607.1 use to every room; do not copy flattened 2 / 5 tokens as design-release values	Verify source
Table 1607.1 Item 27	Multifamily private vs public	Flattened residential block. Hotels and multifamily dwellings list private rooms and corridors serving them, and public rooms and corridors serving them. Concatenated uniform-load cells are not adopted	Group R-2 dwelling units and their corridors / public rooms	One- and two-family attic rows are a different residential block	Direct	After published-table check, load units and unit corridors on the private-room row and lobbies/amenities on the public-room row	Verify source
Table 1607.1 Items 5, 8, 28, 32	Balconies, corridors, roofs, stairs	Flattened cells for balconies/decks (1.5 times served area, cap token unreadable as SI), first-floor corridors, ordinary/occupiable/vegetative roofs, and stairs (one- and two-family vs all other). No reconstructed cell is adopted	Those uses	Footnotes a/b/c control live-load reduction eligibility	Direct	Map unit balconies, tower stairs, ordinary roof and any occupiable roof to the published rows before issuing loads	Verify source
1607.4	Concentrated-load patch	Apply Table 1607.1 concentrated load or the uniform load, whichever governs. Unless otherwise specified, the concentration is uniform over 750 mm by 750 mm at the location producing maximum effects	Floors, roofs and similar surfaces	Concurrent uniform + concentrated is not required	Direct	Check slabs and local framing for the 750 mm patch as well as the uniform case	Verified
1607.5	Movable-partition allowance	Where partition locations are subject to change, include partition weight unless specified live load is 4 kN/m² or greater. Partition live load not less than 0.75 kN/m²	Office buildings and other buildings with movable partitions	Permanent partitions are dead load (Chapter 2 definition, named in commentary only — not used as a value)	Conditional	Apply 0.75 kN/m² on flexible amenity/office floors with live load < 4 kN/m² ; confirm whether unit fit-out partitions are treated as movable	Verified
1607.21	Stair-tread concentration	Table 1607.1 stair concentrated load acts on 50 mm by 50 mm and need not act concurrently with the uniform stair load	Stair treads	Uniform stair load remains a Table 1607.1 OCR cell	Direct	Check tread plates for the 50 mm patch; keep uniform stair load on the published “all other” row	Verified
1607.22.1–1607.22.2	Uninhabitable attic geometry	Without storage: clear height joist-to-rafter < 1050 mm , or trusses that cannot contain a 1050 mm × 600 mm rectangle. With storage: height ≥ 1050 mm or that rectangle fits; load applies only where access opening is ≥ 500 mm × 750 mm at ≥ 750 mm clear height and joist/truss slope is not greater than 2 in 12 ; remaining chords 0.50 kN/m² concurrent	Residential occupancies with attics	Table 1607.1 attic uniform values remain OCR. 1607.22.3: attics served by stairs other than pull-down use habitable-attic/sleeping live load	Conditional	Typical high-rise R-2 has no attic; if a pitched penthouse attic exists, classify storage vs no-storage with these millimetre tests	Verified

10. Guards, grab bars and vehicle barriers

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1607.9.1	Handrail and guard linear load	Design to resist 0.75 kN/m in accordance with SBC 301 §4.5.1. Glass assemblies also comply with Section 2407	Handrails and guards	(1) One- and two-family dwellings: concentrated load only. (2) Group I-3, F, H and S, not public, occupant load < 50: 0.3 kN/m . Neither exception is this R-2 tower	Direct	Design unit and common guards for 0.75 kN/m ; do not use the I-3/F/H/S reduction	Verified
1607.9.1.1	Handrail/guard concentrated load	Resist 0.9 kN concentrated in accordance with SBC 301 §4.5.1 (separate load case)	Handrails and guards	Not combined with the linear load	Direct	Check posts and connections for 0.9 kN at the worst point	Verified
1607.9.1.2	Guard infill load	Balusters, panel fillers and infill (except handrail and top rail) resist 0.2 kN concentrated per SBC 301 §4.5.1.2	Guard infill	Separate load case	Direct	Specify infill/glass clips for 0.2 kN	Verified
1607.9.2	Grab bar / shower seat / accessible bench	Resist a single concentrated 1.1 kN in any direction at any point producing maximum effects	Grab bars, shower seats and accessible benches	None stated	Direct	Coordinate accessible-unit and common-toilet accessories with 1.1 kN fixings into structure	Verified
1607.10	Passenger vehicle barrier	Resist 27 kN concentrated per SBC 301 §4.5.3. Truck/bus garages use an approved traffic-railing method	Vehicle barriers for passenger vehicles	Heavy-vehicle barriers are not this 27 kN case	Conditional	If podium/basement parking has a drop, design the barrier for 27 kN ; do not import bumper height from commentary figures	Verified

11. Passenger garages and heavy vehicles

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1607.7	Passenger-garage concentration	Design garage floors for Table 1607.1 uniform load or 13.5 kN on 115 mm × 115 mm for passenger vehicles not more than nine passengers; mechanical parking without slab/deck: 10 kN per wheel	Passenger-vehicle storage	Table 1607.1 garage uniform cell is OCR. Live-load reduction: 1607.12.1.3 / 1607.12.2 Item 2	Conditional	For a resident podium garage, check the 13.5 kN patch and the published uniform row	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1607.8 / 1607.8.1	Heavy-vehicle access	Surfaces intended for vehicles greater than 45 kN GVWR, or unrestricted access for such vehicles, shall use the jurisdiction's roadway/bridge vehicular live loads including impact and fatigue	Fire-truck lanes, loading courts, heavy access	Passenger-garage loads do not cover this case	Conditional	If SCD appliances or trucks enter a structure, obtain AHJ highway loading; do not use passenger-garage numbers	Verified
1607.8.3	Heavy-vehicle garage	Garages for vehicles exceeding 4500 kg GVWR use 1607.8.1 loading; impact and fatigue not required for those garages	Heavy-vehicle garages	Exception: actual vehicle weights if approved, but not less than 3 kN/m² , non-reducible	Conditional	Do not apply passenger 13.5 kN to a truck basement	Verified
1607.8.4.1	Forklift impact	Increase vehicle and wheel loads by 30 percent for impact where forklifts or movable equipment are intended	Forklifts / movable equipment	Post maximum weight (1607.8.5 / 106.1)	Conditional	If a loading dock or storage uses forklifts, apply the 30 percent impact increase and post the limit	Verified
1607.12.1.3	Garage live-load reduction	Passenger-vehicle garage live loads shall not be reduced, except members supporting two or more floors may be reduced by a maximum of 20 percent , not less than L from 1607.12.1	Passenger vehicle garages	Alternative method 1607.12.2 Item 2 is the same 20 percent cap	Conditional	Do not reduce garage slab/beam live load; column stacks of two or more floors may take 20 percent	Verified

12. Impact, elevators, façade access and fall arrest

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1607.11.1	Elevator dynamic loads	Members subject to elevator dynamic loads shall be designed for impact loads and deflection limits of ASME A17.1	Elevator supports	ASME values are not imported	External verification	Note ASME A17.1 on the lift-shaft structural criteria; do not pick an impact factor from this chapter	Verified
1607.11.2	Machinery impact allowance	Increase machinery weight 20 percent for light shaft- or motor-driven machinery; 50 percent for reciprocating or power-driven units; increase further if the manufacturer specifies	Machinery and moving loads	None stated beyond manufacturer increases	Direct	Apply 20 percent or 50 percent to plant-room equipment loads on the structural schedule	Verified
1607.11.3	Façade-hoist support	In addition to other live loads, elements supporting façade-access/building-maintenance hoists shall be designed for 2.5 times the rated hoist load or the stall load, whichever is larger	BMU davits, hoist beams, façade-access supports	None stated	Direct	Size davit bases, sockets and supporting members for 2.5× rated/stall; show the load path into the frame	Verified
1607.11.4	Fall-arrest / lifeline anchorage	In addition to other live loads, fall-arrest, lifeline and rope-descent anchorages and supporting elements shall be designed for not less than 13.8 kN per attached line, in any direction the load can be applied. Horizontal-lifeline supports shall take the maximum tension from those live loads	Rope-access / fall-arrest anchors	None stated	Direct	Provide certified 13.8 kN anchors and a continuous structural path; do not treat façade brackets as architectural-only	Verified

13. Live-load reduction, roofs and interior walls

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1607.12.1	Basic floor live-load reduction	Where KLL AT ≥ 37 m² , reduce per Equation 16-6: L = Lo (0.25 + 4.57 / √(KLL AT)) . L not less than 0.50 Lo for members supporting one floor; not less than 0.40 Lo for two or more floors	Uniform floor live loads except roofs	Subject to 1607.12.1.1–1607.12.1.3 and Table 1607.1 footnotes. Table 1607.12.1 KLL is flattened OCR	Direct	SER may reduce residential floor live load only after a published KLL check; keep one-floor members ≥ 0.50 Lo	Verified
1607.12.1.1	One-way slab tributary cap	AT for Equation 16-6 shall not exceed slab span times 1.5 times the slab span	One-way slabs	None stated	Conditional	If one-way slabs are used, cap the reduction area at 1.5 span width	Verified
1607.12.1.2	Heavy live-load reduction ban	Live loads exceeding 5 kN/m² shall not be reduced	Heavy uniform live load	(1) Members supporting two or more floors: max 20 percent , not less than 1607.12.1 L. (2) Non-storage, approved rational reduction	Conditional	Do not reduce storage or other > 5 kN/m² floors except the stacked-member 20 percent	Verified
1607.12.2	Alternative reduction	Alternative to 1607.12.1: no reduction where L > 5 kN/m² except 20 percent on two-or-more-floor members; same garage rule; for L ≤ 5 kN/m² and supported area ≥ 14 m² , R = 0.861(A – 13.94) percent, not exceeding 40 percent (one floor), 60 percent (two or more), or R = 23.1(1 + D/Lo) . One-way slab A capped at 0.5 times span width	Designer elects the alternative	Table 1607.1 limitations still apply	Conditional	If this method is used, apply the 14 m² start and the 40 / 60 percent caps	Verified
1607.14.2.1	Ordinary roof live-load reduction	Ordinary flat/pitched/curved roofs and non-fabric awnings/canopies may use Lr = Lo R1 R2 with 0.6 ≤ Lr ≤ 1 (kN/m²). R1 = 1 for AT ≤ 18 m² ; 1.2 – 0.011 AT between 18 and 54 m² ; 0.6 for AT ≥ 54 m² . R2 from slope factor F (0.12 × slope percent, or rise-to-span × 32). Greenhouses: not less than 0.6 kN/m²	Ordinary (non-occupiable) roofs	Occupiable roofs use 1607.14.2.2 / 1607.12 instead	Direct	For a non-occupiable tower roof, apply Equation 16-9; do not take Lr below 0.6 kN/m²	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1607.14.2.2	Occupiable roof live load	Occupiable roofs (vegetative, landscaped, assembly or similar) and marquees may reduce uniform live load per 1607.12 (floor method), not 1607.14.2.1	Occupiable roofs	Assembly live-load reduction still follows Table 1607.1 footnotes / 1607.12 limits	Conditional	If the roof is an amenity terrace, use the occupancy-served live load (published Table 1607.1) and floor reduction rules	Verified
1607.14.4.1	PV roof live-load cases	Design the roof with PV dead load and without PV. Exception: roof live load need not be applied to the area covered by panels where clear space between panels and roof is 600 mm or less	Roofs supporting PV	Ground-mounted inaccessible arrays need not take roof PV live load (1607.14.4.4)	Conditional	If PV is installed, run both presence cases; omit live load only under panels with ≤ 600 mm clear	Verified
1607.14.4.3	Open-grid PV roof	Open-grid framing without deck/sheathing: uniform roof live load may be reduced to 0.57 kN/m²	Open-grid PV support	Concentrated loads of 1607.14.4.1 still apply	Conditional	Use 0.57 kN/m² only on true open-grid PV structures, not on a concrete podium roof	Verified
1607.16	Interior wall horizontal load	Interior walls and partitions exceeding 1.8 m in height, including finishes, shall resist the loads to which they are subjected but not less than 0.25 kN/m² horizontal	Interior walls / partitions > 1.8 m	Fabric partitions: 1607.16.1. Fire walls: 1607.16.2	Direct	Design unit demising and corridor partitions above 1.8 m for 0.25 kN/m²	Verified
1607.16.1	Fabric-partition loads	Fabric partitions > 1.8 m : horizontal distributed load on framing from the fabric face area, plus 0.2 kN on a 200 mm -diameter (31415 mm²) patch at 1.4 m above the floor	Fabric partitions	Alternative to the 1607.16 uniform wall load	Conditional	If fabric office partitions are used in amenities, apply the 0.2 kN / 1.4 m patch	Verified
1607.16.2	Fire-wall residual load	Fire walls and supports shall withstand a minimum horizontal allowable-stress load of 0.25 kN/m² where the structure on either side has collapsed (Section 706.2 stability)	Fire walls	None stated	Conditional	If a fire wall is used at a podium separation, design it to stand with 0.25 kN/m² after one-side collapse	Verified

14. Wind

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1609.1.1	Wind-load determination	Determine wind loads per SBC 301 Chapters 26 to 30. Opening protection, V and exposure may be determined per 1609 or SBC 301. Wind from any horizontal direction, pressures normal to the surface	Every building	(1) ICC 600 for applicable Group R-2/R-3 within 1609.1.1.1 limits. (2) AWC WFCM. (3) AISI S230. (4) NAAMM FP 1001. (5) TIA-222 with topographic extent 16 times escarpment height. (6) Wind tunnel ASCE 49 / SBC 301 §§31.4–31.5. ICC 600 / WFCM / AISI S230 are not typical for a high-rise	External verification	Design the tower to SBC 301 Ch. 26–30; do not use ICC 600 prescriptive tables on this high-rise	Verified
1609.1.1.1	Prescriptive-wind site limits	ICC 600 only in Exposure B or C. ICC 600, WFCM and AISI S230 shall not apply on the upper half of an isolated hill/ridge/escarpment that is 18 m or higher (Exposure B) or 9 m (Exposure C), with max average slope > 10 percent , unobstructed upwind for 50 times the height or 1.6 km , whichever is greater	Attempted use of those standards	High-rise R-2 remains on SBC 301	Not typical	Do not open an ICC 600 path for this tower	Verified
1609.2	Wind-borne debris glazing	In wind-borne debris regions, glazing shall be impact-resistant or protected: large-missile ASTM E1996 within 9 m of grade; small-missile more than 9 m above grade	Wind-borne debris regions (definition is Chapter 2 / 1609, not quantified here)	(1) Risk Category I may be unprotected. (2) Risk Category II–IV glazing over 18 m above ground and over 9 m above aggregate-surfaced roofs within 450 m may be unprotected	Conditional	Confirm whether the site is a wind-borne debris region before specifying missile-rated glass; Exception 2 may cover typical tower glazing above 18 m	Verified
1609.2.1	Louver missile test	Louvers protecting intake/exhaust ducts not assumed open, within 9 m of grade, shall meet AMCA 540	Those louvers in wind-borne debris regions	None stated	Conditional	Specify AMCA 540 louvers on low-level intakes if the debris-region trigger applies	Verified
1609.3	Basic design wind speed	V (m/s) from Figures 1609.3(1)–(4) by risk category. Mountainous terrain, gorges and special wind regions shall be examined; the building official may adjust using SBC 301 Chapter 26	Wind design	Figures are not readable as numeric speeds in this extract	External verification	SER to read the published KSA map for the locked risk category; do not invent a speed	Verified
1609.3.1 / Table 1609.3.1	ASD wind-speed conversion	Convert mapped V to Vasd with Table 1609.3.1 or Vasd = V √0.6 . Table cells are concatenated OCR and are not adopted	Where Vasd is required (1609.1.1 Exceptions 4 and 5 methods)	Linear interpolation permitted (table Note a)	External verification	Prefer Equation 16-16; do not read flattened table pairs as design speeds	Verify source
1609.4.1–1609.4.3	Exposure category	For each direction, determine exposure in two upwind 45-degree sectors and use the worse. Exposure B: mean roof ≤ 9 m needs 450 m of Surface Roughness B; mean roof > 9 m needs 780 m or 20 times building height, whichever is greater. Exposure D: Surface Roughness D for 1500 m or 20h , and sites within 180 m or 20h of that D fetch. Exposure C is all other cases. Surface Roughness C includes scattered obstructions generally < 10 m	Wind exposure	High-rise mean roof height is > 9 m , so Exposure B needs the 780 m / 20h fetch	Direct	Classify urban Riyadh fetch with the > 9 m Exposure B test; default to C if the B fetch is not proven	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1609.5.1–1609.5.2	Roof deck and covering	Roof deck shall withstand SBC 301 wind pressures; roof coverings shall comply with 1609.5.1. Asphalt shingles over a complying deck also comply with 1504.2	Roof systems	Air-permeable rigid tile may use 1609.5.3	External verification	Specify deck and covering wind resistance from SBC 301 / Chapter 15; do not import pressures here	Verified
1609.5.3	Rigid-tile uplift moment	Ma = qh CL b L La [1.0 – GCp] / 1000. CL = 0.2 (or 1504.3.1 test). Point of uplift at 0.76L . Permitted only for tiles with head lap ≥ 50 mm , length 300–525 mm , exposed width 200–375 mm , tail thickness ≤ 30 mm , and mortar/adhesive contact not more than two-thirds of tile area	Concrete/clay rigid tile roofs	GCp from SBC 301 Chapter 30, not adjusted for internal pressure	Conditional	If clay/concrete tile is used, stay inside the millimetre envelope and compute Ma ; otherwise use 1609.5.1	Verified

15. Soil, hydrostatic and rain

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1610.1 / Table 1610.1	Lateral soil load	Foundation and retaining walls shall resist Table 1610.1 minimum lateral soil loads unless a geotechnical investigation per SBC 303 determines otherwise. Restricted-top walls use at-rest pressure; free-to-rotate retaining walls may use active. Add surcharge; increase for expansive soils. Undrained backfill: full hydrostatic unless drainage per SBC 303 §§13.4.3–13.4.4. Table cells are concatenated OCR	Foundation / retaining walls	Foundation walls not more than 2.4 m below grade and laterally supported at the top by flexible diaphragms may use active pressure	Direct	Prefer SBC 303 pressures; if Table 1610.1 is used, verify published active/at-rest cells. Apply the 2.4 m active-pressure exception only to qualifying basement walls	Verify source
1610.2	Uplift on below-grade floors	Basement floors, slabs on ground and similar elements below grade shall resist uplift. Upward water pressure is full hydrostatic over the entire area, measured from the underside. Expansive-soil uplift complies with SBC 303	Below-grade floors where water/expansive soil applies	None stated beyond SBC 303	Conditional	If a basement exists, design the slab for hydrostatic uplift from the underside and provide drainage or thickness/anchorage	Verified
1611.1	Rain load	Design each roof portion for rainwater per SBC 301 Chapter 8. Design rainfall is the 100-year 15-minute event, other approved local rates, or twice the 100-year hourly rate in Figure 1611.1. R = 0.0098 (ds + dh) with ds, dh in mm and R in kN/m² on the undeflected roof	Roofs	Figure 1611.1 rainfall map is not readable as a numeric rate here. Commentary worked example is not adopted	External verification	Size primary/secondary drainage (SBC 701) then compute R from static + hydraulic head; do not import the commentary 2.74 kN/m² example	Verified
1611.2	Ponding instability	Evaluate susceptible bays for ponding instability per SBC 301 Chapters 7 and 8	Roofs with susceptible bays	Assume primary drains blocked	Direct	Provide roof slope/camber and stiffness so rain cannot progressive-pond	External verification
1611.3	Controlled drainage	Roofs with flow-control hardware shall have a higher secondary system; design for water to the secondary elevation plus the 1611.1 hydraulic head, and check 1611.2 ponding	Controlled-flow roof drains	None stated	Conditional	If controlled-flow drains are used, set the overflow elevation and the corresponding rain load	Verified

16. Flood, snow, ice and tsunامي

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1603.1.7	Flood data on CDs	If the building is wholly or partly in a flood hazard area (SBC 301 §5.1), show lowest-floor elevation (including basement) and, in high-velocity wave areas, the bottom of the lowest horizontal structural member	Flood hazard areas	Non-residential dry floodproofing elevation is listed for non-wave zones	Conditional	Confirm the AHJ flood map; if not in a flood hazard area, omit 1603.1.7 data	Verified
1612.1	Flood loads	Design in flood hazard areas, coastal high-hazard areas and coastal A zones per SBC 301 Chapter 5	Those mapped areas	Values not in this chapter	External verification	Do not import flood elevations from SBC 301 here	Verified
1614.1	Atmospheric ice	Atmospheric ice load is inapplicable to KSA	KSA projects	None stated	Direct	Omit ice accretion from the load-criteria sheet	Verified
1615.1	Tsunami loads	Risk Category III and IV buildings in Tsunami Design Zones (Tsunami Design Geodatabase) shall comply with SBC 301 Chapter 6 as modified	RC III/IV in a mapped tsunami zone	Not an inland Riyadh R-2 default	Not typical	Do not apply tsunami detailing unless the AHJ maps a zone and risk category is III/IV	External verification

17. Earthquake

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1613.1	Seismic scope	Every structure, including permanently attached nonstructural components and their supports and attachments, shall resist earthquake motions per SBC 301 Chapters 11, 12, 13, 15, 17 and 18 as applicable. SDC may be determined per 1613 or SBC 301	All structures	(1) Detached one- and two-family in SDC A/B/C or Ss < 0.4g . (2) Agricultural storage, incidental occupancy. (3) Special structures under other regulations. (4) SBC 301 Chapter 14 references shall not apply except as specifically required. Exception 1 is not this high-rise	External verification	Design the tower and permanently attached façades/MEP per SBC 301; do not use the dwelling exception	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1613.2.1	Mapped accelerations / SDC A shortcut	Determine Ss and S1 from Figures 1613.2.1(1)–(2). Where S1 ≤ 0.04g and Ss ≤ 0.15g , SDC A is permitted	Site seismicity	Figures are not readable as values in this extract	External verification	SER to read published maps; do not assume SDC A for Riyadh	Verified
1613.2.2	Site class default	Classify the site as A–F per SBC 301 Chapter 20. If soil properties are not known in sufficient detail, use Site Class D , unless the building official or geotechnical data determine Class E or F. Class B rock without site-specific velocity: Fa and Fv = 1.0	Site classification	Measurements over the top 30 m	Direct	Until a geotechnical report exists, the code default is D , not A/B	Verified
1613.2.3 / Tables 1613.2.3(1)–(2)	Site coefficients	SMS = Fa Ss and SM1 = Fv S1 , but SMS shall not be taken less than SM1 except when determining SDC per 1613.2.5. Default Site Class D: Fa not less than 1.2 . Tables of Fa / Fv are concatenated OCR (the second table is titled Fa in the extract but tabulated vs S1). No reconstructed coefficient is adopted	Seismic ground-motion adjustment	Site Class F and some E cells are “Note b” → SBC 301 §11.4.8	External verification	Compute SMS / SM1 from published Fa / Fv ; do not use flattened 1.6 / 2.4 tokens	Verify source
1613.2.4	Design spectral accelerations	SDS = (2/3) SMS ; SD1 = (2/3) SM1	After site-adjusted MCER	None stated	Direct	Reduce site-adjusted accelerations by 2/3 for design	Verified
1613.2.5 / Tables 1613.3.5(1)–(2)	Seismic design category	If S1 ≥ 0.75g : Risk Category I–III → SDC E ; Risk Category IV → SDC F . Otherwise assign the more severe category from Tables 1613.3.5(1) and (2) using SDS , SD1 and risk category, irrespective of period T. Table range cells (0.167g / 0.33g / 0.50g and 0.067g / 0.133g / 0.20g) are concatenated OCR and are not adopted	SDC assignment	Alternative short-period determination 1613.2.5.1 where S1 < 0.75g and four listed conditions (including diaphragm spacing ≤ 12 m)	External verification	Assign SDC from published tables after SDS / SD1 are known; do not reconstruct the flattened A/B/C/D grid	Verify source
1613.3	Ballasted rooftop PV	Ballasted nonpenetrating PV need not be rigidly attached. Allowed only on roofs not steeper than 1:12 . Resist sliding/uplift per 1605 using an engineered friction coefficient. In SDC C–F, accommodate seismic displacement by nonlinear RHA, approved analysis or shake-table testing consistent with SBC 301 roof nonstructural forces	Ballasted nonpenetrating PV	Penetrating/attached arrays are not this exception	Conditional	If ballasted PV is used, keep roof slope ≤ 1:12 and run a sliding/displacement check	Verified

18. Structural integrity

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1616.1	Integrity charging trigger	High-rise buildings assigned to Risk Category III or IV shall comply with 1616.2 (frame) or 1616.3 (bearing wall)	High-rise and RC III or IV	Risk Category II high-rise is not charged by 1616.1	Conditional	Apply 1616 only if risk category is III or IV; a typical RC II apartment tower is outside this section	Verified
1616.2.1	Concrete-frame integrity steel	Concrete frames conform to SBC 304 §4.10. Where that standard requires steel through the column cage, nominal tensile strength is two-thirds of the required one-way vertical strength of the floor/roof-to-column connection in each direction	RC III/IV high-rise concrete frames	Continuous slabs with reinforcement ≥ 0.0015 times concrete area in each orthogonal direction, monolithic or equivalently bonded: tensile strength one-third	Conditional	If 1616 applies, detail through-column integrity steel to 2/3 (or 1/3 with the slab exception)	Verified
1616.2.2.1	Steel-frame column splice tension	Each column splice shall have tensile design strength to transfer design dead and live load tributary to the column between that splice and the splice or base immediately below	RC III/IV high-rise steel / joist / composite frames	None stated	Conditional	If 1616 applies, check every column splice for the storey tributary D+L tension	Verified
1616.2.2.2	Steel-frame beam end ties	Beam/girder end connections: nominal axial tensile strength equal to required vertical shear (ASD) or two-thirds of required shear (LRFD), but not less than 45 kN . Shear and axial tension need not act together	Those frames	Concrete slab or slab-on-deck with headed studs ≥ 10 mm diameter at ≤ 300 mm o.c. average, plus slab reinforcement ≥ 0.0015 in two directions: half (ASD) or one-third (LRFD) of required shear, still ≥ 45 kN	Conditional	If 1616 applies, provide beam-end tensile capacity ≥ 45 kN (or the reduced slab exception)	Verified
1616.3.2.1–1616.3.2.2	Bearing-wall longitudinal / transverse ties	Other-than-precast-concrete bearing-wall structures: longitudinal ties across interior load-bearing walls at not more than 3 m centres; transverse ties at not more than the bearing-wall spacing. TT = w L S ≤ αT S with αT = 22 kN/m (masonry) or 5.5 kN/m (cold-formed steel). ASD tensile strength may be 1.5 times allowable stress times area	RC III/IV high-rise bearing-wall structures not under 1616.3.1	Precast concrete walls use SBC 304 §§16.2.4–16.2.5 (1616.3.1)	Conditional	If a bearing-wall III/IV high-rise is used, space longitudinal ties at ≤ 3 m and size TT	Verified
1616.3.2.3	Bearing-wall perimeter ties	Perimeter ties within 1.2 m of the edge of each floor and roof; Tp = 90.7 w ≤ βT with βT = 71 kN (masonry) or 17.8 kN (CFS). ASD may use 1.5 times allowable stress times area	Same bearing-wall trigger	None stated	Conditional	Keep perimeter ties inside the 1.2 m edge band	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1616.3.2.4	Bearing-wall vertical ties	Continuous vertical ties over the building height; capacity not less than wall weight in the storey plus diaphragm tributary to the wall in the storey below. Not fewer than two ties per wall. Each tie need not exceed 44 kN/m (masonry) or 11 kN/m (CFS) of wall tributary to the tie	Same bearing-wall trigger	None stated	Conditional	Provide at least two continuous vertical ties per bearing wall	Verified

19. Project-use controls

1. Use **Verified** rows for coordination after the row trigger and branch are confirmed.
2. Treat every **Verify source** row (all reconstructed appended tables, flattened **KLL / Fa / Fv** / SDC grids, and the two helipad uniform-load tokens) as a design hold point; no affected value is to be placed on issued-for-approval drawings without a published-source check.
3. Do not import mapped wind speeds, **Ss / S1**, live-load occupancy cells, or deflection L/n ratios from memory, CS.md, or commentary examples.
4. Do not apply ICC 600 / AWC WFCM / AISI S230 prescriptive wind provisions to this high-rise.
5. Do not apply Section 1616 integrity ties unless the building is high-rise **and** Risk Category III or IV.
6. Do not apply one- and two-family attic, dwelling-stair, or guard exceptions to this Group R-2 tower.
7. Record risk category, site class, exposure, roof use and garage class in the project Golden Thread; this matrix is not evidence of SCD NOC or stamped structural compliance.

20. Coverage summary

Internal inventory of the attached Chapter 16 extract (numbered code, exceptions, tables, footnotes; commentary excluded). Row-level records are not published.

- **Inventory scope:** numbered code, exceptions, tables, footnotes (commentary excluded)
- **Total independently checkable numeric records:** 437
- **Verified:** 191
- **Verify source:** 246

Counts by top-level section

Top-level section	Records
1601	0
1602	0
1603	0
1604	45
1605	9
1606	0
1607	209
1608	0
1609	45
1610	25
1611	3
1612	0
1613	77
1614	0
1615	0
1616	24

Appended-table coverage

Appended table	Records	Verify source records
Table 1604.3 (incl. footnotes a, d, f, h, i)	29	29
Figure/Table 1604.3.6 (concatenated after SDC tables)	4	4
Table 1604.5 (numeric occupancy thresholds only)	7	7
Table 1607.1	98	98
Table 1607.12.1	7	7
Table 1609.3.1	12	12
Table 1610.1	24	24
Table 1613.2.3(1)	27	27
Table 1613.2.3(2)	30	30

Appended table	Records	Verify source records
Table 1613.3.5(1)	3	3
Table 1613.3.5(2)	3	3

Helipad uniform-load tokens in 1607.6 Items 1.1/1.2 (1.12 / 1.23 concatenated with list markers) are **2** additional Verify source records, counted under Section 1607.

Coverage cross-check against SBC 201 Chapter 16 Structural Design (2024)_CS.md was topics-only: SBC 301 companions; risk category; mixed occupancy. No CS.md value was copied into a matrix cell.

21. Unresolved-source register

Hold points for the 246 **Verify source** inventory records. Counts are record counts, not distinct numeric values. No value in this register is a design-release figure.

Affected table / clause	Why unverified	Control note
Table 1604.3 + footnotes	29 flattened L/n and footnote factors (L/60, 0.42, 0.5D, aluminium L/175, etc.)	Verify published table before specifying finish-dependent deflection; do not adopt concatenated L/360 tokens
Figure/Table 1604.3.6	4 concatenated concrete-roof deflection cells mixed into the SDC page	Verify published Figure 1604.3.6; outbound SBC 304 still governs concrete
Table 1604.5	7 flattened occupant-load / care thresholds (300, 2,500, 250, 500, 50, 5,000 and related)	Conventional R-2 is typically II; verify published III/IV triggers before raising category
Table 1607.1	98 flattened uniform/concentrated occupancy cells across four page-splits (tokens such as 2.59, 59, 44.5)	After published-table check, use multifamily private/public, balcony, stair, garage and roof rows; omit unrelated occupancies
Table 1607.12.1	7 flattened KLL element factors	Do not reduce live load until published KLL values are confirmed
1607.6 Items 1.1–1.2	2 concatenated list-marker + kN/m² tokens (1.12, 1.23)	Helipads are not typical; verify published uniform helipad loads if a pad is added
Table 1609.3.1	12 flattened V / Vasd pairs	Prefer Equation 16-16; do not read concatenated speed pairs
Table 1610.1	24 flattened active/at-rest soil pressures	Prefer SBC 303 investigation; verify published cells if the table is used
Tables 1613.2.3(1)–(2)	57 flattened Fa / Fv cells (second table title OCR as Fa)	Verify published coefficients; default Class D still requires Fa ≥ 1.2 from code text
Tables 1613.3.5(1)–(2)	6 flattened SDC range bounds	Verify published SDS / SD1 grids before assigning SDC A–D